COURSE SYLLABUS

BOROUGH OF MANHATTAN COMMUNITY COLLEGE

The City University of New York
Department of Computer Information Systems

Title of Course: **Python Programming for Data Science** — **Class Hours:** 2
CSC 203 – 090L — **Laboratory Hours:** 2
Spring 2026 — **Credits: 3**

Instructor: Younes Benkarroum
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Office Hours: Monday 2:00pm – 4:00pm
Office Number: F-1030L (Phone Extension 7227)

Course Description:
This course builds upon Python programming skills acquired in CSC 101. Students will be introduced to data science tools and advanced Python programming topics such as Dictionaries and Sets, Array-Oriented Programming with NumPy, Strings, Files and Exceptions, Object-Oriented Programming, Recursion, Searching, Sorting and Big O.
By the end of this course, students are expected to become proficient in Python programming, master the basic methods of data visualization, and be able to manipulate multidimensional data at an advanced level using data science libraries.

Prerequisites:
(CSC 101 or CSC 103) and (MAT 206 or MAT 200), or departmental approval.

Course Student Learning Outcomes (Students will be able to…)	Measurements (means of assessment for student learning outcomes listed in first column)
1. Use sets and dictionaries to represent unordered collections of unique values and key–value pairs.	**1.** Lab exercises and exam questions.
2. Design and develop array-oriented program using the numerical Python library NumPy.	**2.** Programming projects and lab exercises.
3. Implement exception detection and handling to avoid the termination of a program.	**3.** Programming projects and exam questions.
4. Apply object-oriented programming (OOP) techniques to develop applications.	**4.** Programming projects, lab exercises and exam questions.
5. Use Big O notation to compare the efficiencies of searching and sorting algorithms.	**5.** Lab exercises and exam questions.

Below are the college's general education learning outcomes, the outcomes that are checked in the left-hand column indicate goals that will be covered and assessed in this course. (Check at least one.)

	General Education Learning Outcomes	Measurements (means of assessment for general education goals listed in first column)
☑	**Communication Skills-** Students will be able to write, read, listen and speak critically and effectively.	Program documentation and project presentation.
☑	**Quantitative Reasoning-** Students will be able to use quantitative skills and the concepts and methods of mathematics to solve problems.	Use formulas and concepts of mathematics to analyze data and solve problems in programming assignments.
☐	**Scientific Reasoning-** Students will be able to apply the concepts and methods of the natural sciences.	
☐	**Social and Behavioral Sciences-** Students will be able to apply the concepts and methods of the social sciences.	

☐	**Arts & Humanities-** Students will be able to develop knowledge and understanding of the arts and literature through critiques of works of art, music, theatre or literature.	
☑	**Information & Technology Literacy-** Students will be able to collect, evaluate and interpret information and effectively use information technologies.	Use a high-level computer programming language to create application software.
☐	**Values-** Students will be able to make informed choices based on an understanding of personal values, human diversity, multicultural awareness and social responsibility.	

Required Text & Readings:

Textbook: **Intro to Python for Computer Science and Data Science, 1st Edition**
Author: **Paul Deitel and Harvey Deitel**
Publisher: **Pearson**
ISBN: **978-0-13-5404676**

Required software:

Anaconda Python distribution (free). The software includes most of the Python, visualization and data science libraries needed for this course, as well as Python, the IPython interpreter, Jupyter Notebooks and Spyder.

Other Resources:

Flash drives are recommended.

Evaluation & Requirements of Students:

Midterm	30%
Final	30%
Quizzes	20%
Lab Tasks	20%
Total:	**100%**

Outline of Topics:

Week No	Chapter	Topic
Week 1	**Chapter 2**	**CSC 101 Review: Introduction to Python Programming**
Week 2	**Chapter 3**	**CSC 101 Review: Control Statements and Program Development**
Week 3	**Chapter 4**	**Functions: Measures of Dispersion**
Week 4	**Chapter 5**	**Lists and Tuples: Simulation and Static Visualizations**
Week 5	**Chapter 6**	**Dictionaries and Sets: Dynamic Visualizations**
Week 6	**Chapter 7**	**Array-Oriented Programming with NumPy**
Week 7	**Chapter 7**	**Array-Oriented Programming with NumPy**
Week 8		**Review & Midterm Exam**
Week 9	**Chapter 8**	**Strings: Pandas, Regular Expressions and Data Munging**
Week 10	**Chapter 9**	**Files and Exceptions: Working with CSV Files**
Week 11	**Chapter 10**	**Object-Oriented Programming and its Application in Data Science**
Week 12	**Chapter 10**	**Object-Oriented Programming and its Application in Data Science**
Week 13	**Chapter 11**	**Recursion, Searching, Sorting, Big O, and Visualizing Algorithms**
Week 14	**Chapter 11**	**Recursion, Searching, Sorting, Big O, and Visualizing Algorithms**
Week 15		**Final Exam**

Class Participation
Participation in the academic activity of each course is a significant component of the learning process and plays a major role in determining overall student academic achievement. Academic activities may include, but are not limited to, attending class, submitting assignments, engaging in in-class or online activities, taking exams, and/or participating in group work. Each instructor has the right to establish their own class participation policy, and it is each student's responsibility to be familiar with and follow the participation policies for each course.
BMCC is committed to the health and well-being of all students. It is common for everyone to seek assistance at some point in their life, and there are free and confidential services on campus that can help.

Single Stop www.bmcc.cuny.edu/singlestop, room S230, 212-220-8195, singlestop@bmcc.cuny.edu. If you are having problems with food or housing insecurity, finances, health insurance or anything else that might get in the way of your studies at BMCC, come by the Single Stop Office for advice and assistance. Assistance is also available through the Office of Student Affairs, S350, 212-220-8130, studentaffairs@bmcc.cuny.edu.

Counseling Center www.bmcc.cuny.edu/counseling, room S343, 212-220-8140, counselingcenter@bmcc.cuny.edu. Counselors assist students in addressing psychological and adjustment issues (i.e., depression, anxiety, and relationships) and can help with stress, time management and more. Counselors are available for walk-in visits.

Office of Compliance and Diversity https://www.bmcc.cuny.edu/about-bmcc/compliance-diversity, room S701, 212-220-1236. BMCC is committed to promoting a diverse and inclusive learning environment free of unlawful discrimination/harassment, including sexual harassment, where all students are treated fairly. For information about BMCC's policies and resources, or to request additional assistance in this area, please visit or call the office, or email olevy@bmcc.cuny.edu, or twade@bmcc.cuny.edu. If you need immediate assistance, please contact BMCC Public safety at 212-220-8080.

Office of Accessibility www.bmcc.cuny.edu/accessibility, room N360 (accessible entrance: 77 Harrison Street), 212-220-8180, accessibility@bmcc.cuny.edu. This office collaborates with students who have documented disabilities, to coordinate support services, reasonable accommodations, and programs that enable equal access to education and college life. To request an accommodation due to a documented disability, please visit, call the office or email.

BMCC Policy on Plagiarism and Academic Integrity Statement
Plagiarism is the presentation of someone else's ideas, words or artistic, scientific, or technical work as one's own creation. Using the idea or work of another is permissible only when the original author is identified. Paraphrasing and summarizing, as well as direct quotations, require citations to the original source. Plagiarism may be intentional or unintentional. Lack of dishonest intent does not necessarily absolve a student of responsibility for plagiarism. Students who are unsure how and when to provide documentation are advised to consult with their instructors. The library has guides designed to help students to appropriately identify a cited work. The full policy can be found on BMCC's Web site, www.bmcc.cuny.edu. For further information on integrity and behavior, please consult the college bulletin (also available online).

Supplemental Instruction (SI)
BMCC is committed to student success; many foundational and gateway courses, including this course, utilize Supplemental Instruction (SI). SI is a free, voluntary academic success program that offers study sessions led by trained SI Leaders. Students are strongly encouraged to attend at least five out-of-class SI sessions during the semester, as institutional data indicate that students who reach this level of participation experience pass rates that are approximately 10–30% higher than those who do not attend SI sessions. Students who attend more than five sessions demonstrate even higher pass rates, making SI a valuable resource for academic success.
SI Leaders are paid student advocates who have mastered the course content and have been trained to facilitate collaborative group sessions. They attend class meetings and schedule two out-of-class small-group sessions per week to review course material, discuss key concepts, develop effective study strategies, and prepare for exams and assignments. Please note: SI Leaders do not grade student work or influence course grades. If you have questions about how SI works in this class, please contact the SI Leader or the professor. The SI Leader for this course is Jafar Abubakar.